Technical Specification

Loan receipt from the Fiori app and a classic driver report

Item	Value
Document	TS-20261105-library-books-loans
App	library-books-loans
Date	05.11.2026
Author	RAP Studio · AI agent
Request	Print the Adobe loan receipt form ZF_LOAN_RECEIPT from the Fiori app and from a classic report as well.
Status	Delivered

1. Purpose and scope

The loan gets a Print Receipt action: it renders the Adobe form ZF_LOAN_RECEIPT (XDP layout, data provider ZFDP_LOAN_RECEIPT) and keeps the PDF in the loan, where the object page offers it as Receipt (PDF). The classic report ZLOAN_RECEIPT_REPORT prints the same form the SAP GUI way with FP_JOB_OPEN, FP_FUNCTION_MODULE_NAME, the form's generated function module and FP_JOB_CLOSE, as a preview or into the output queue. A List Report / Object Page app library-fe shows books, loans and the new action.

2. Requirements

- Print the loan receipt from the Fiori app (button on the loan)
- Print the loan receipt from a classic report with a selection screen
- Use the existing Adobe form ZF_LOAN_RECEIPT unchanged

3. Objects

Object	Type	File	Lines	Description
ZBOOK_A	Table	zbook_a.tabl.cds	16	
ZBP_R_BOOK	Class	zbp_r_book.clas.abap	6	Handler printReceipt: form reader, FDP read_to_xml_v2, render_pdf, stores the PDF
ZBP_R_BOOK	Class (local types)	zbp_r_book.clas.locals_imp.abap	298	Handler printReceipt: form reader, FDP read_to_xml_v2, render_pdf, stores the PDF
ZC_BOOK	Behavior definition	zc_book.bdef.asbdef	25	
ZC_BOOK	CDS view	zc_book.ddls.asddls	27	
ZC_BOOK	Metadata extension	zc_book.ddlx.asddlxs	58	
ZC_LOAN	CDS view	zc_loan.ddls.asddls	34	
ZC_LOAN	Metadata extension	zc_loan.ddlx.asddlxs	58	
ZCL_LIBRARY_CONSOLE	Class	zcl_library_console.clas.abap	115	
ZCL_LOAN_RECEIPT_PRINT	Class	zcl_loan_receipt_print.clas.abap	76	
ZF_LOAN_RECEIPT	Print form	zf_loan_receipt.form.json	6	
ZFDP_LOAN_RECEIPT	Service definition	zfdp_loan_receipt.srvd.srvdsrv	6	
ZI_BOOKLOANSTATS	CDS view	zi_bookloanstats.ddls.asddls	13	

Object	Type	File	Lines	Description
ZI_LOANRECEIPT	CDS view	zi_loanreceipt.ddls.asddls	26	
ZLIBRARY	Message class	zlibrary.msag.json	8	
ZLOAN_A	Table	zloan_a.tabl.cds	16	Loan table: receipt PDF, MIME type and file name
ZLOAN_RECEIPT_REPORT	Program	zloan_receipt_report.prog.abap	120	Classic driver report: member, preview or print; FP_JOB_OPEN … FP_JOB_CLOSE
ZR_BOOK	Behavior definition	zr_book.bdef.asbdef	78	
ZR_BOOK	CDS view	zr_book.ddls.asddls	28	
ZR_LOAN	CDS view	zr_loan.ddls.asddls	24	
ZUI_LIBRARY	Service definition	zui_library.srvd.srvdsrv	6	

4. Selection screen of ZLOAN_RECEIPT_REPORT

Field	Kind	Reference	Text	Mandatory
P_MEMBER	Parameter	c LENGTH 80	Member	
P_PREV	Checkbox		Preview (PDF back, no print)	

5. Design

RAP action printReceipt

For each selected loan the handler reads the form object, the data provider's XML for the loan key and renders the XDP with cl_fp_ads_util=>render_pdf. The PDF is written to the loan with MODIFY ENTITIES IN LOCAL MODE; errors of the form processing are returned as messages in reported.

```
DATA(xml) = fdp_util->read_to_xml_v2( it_select = select_keys ).
cl_fp_ads_util=>render_pdf( EXPORTING iv_xml_data = xml iv_xdp_layout = layout IMPORTING ev_pdf = DATA(pdf) ).
```

Classic driver report

The report reads the member's open loan with its book and passes it as interface parameter LOANRECEIPT to the form's generated function module; the XDP binds $.LoanReceipt.BookTitle and so on. With Preview the PDF comes back in /1BCDWB/FORMOUTPUT, otherwise it goes to the output queue.

- outputparams-getpdf = Preview
- docparams-langu E, country US: en_US

6. Data sources

Tables and views read: zbook_a, zloan_a.

7. Limitations and open points

- The receipt PDF is stored in the loan; printing it again replaces it.
- The report prints the member's open loan with the oldest due date (else their latest loan).

8. Quality

- Code checks (ATC, check variant Default check variant): 0 errors, 0 warnings, 7 infos (ATC Check Report).

- Tests: 5 of 5 passed (QA Test Report).